

Chapter 3 Exercises

Legrand Jones II

1

a

One way we could define $mul_{\mathbb{N}} : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ is by

$$mul_{\mathbb{N}}(m)(0_{\mathbb{N}}) := 0_{\mathbb{N}}$$

$$mul_{\mathbb{N}}(m)(succ_{\mathbb{N}}(n)) := add_{\mathbb{N}}(mul_{\mathbb{N}}(m)(n), m)$$

Conveying the usually expected properties of multiplication (particularly $m(n+1) = mn + m$). We will proceed by defining $m : \mathbb{N} \vdash mul_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N}$ to get our desired term. We will accomplish this by producing our base case and inductive step for multiplication by m and using N-IND.

We (I) want to explicitly describe how to express $p_s(m)$ to apply N-IND. In common language $p_s(m) : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ expresses that if we know how to multiply by m (being a term in our context) on n (our first input) then we should be able to take a term of $P(n)$ (our second input), i.e. an expression of “ $m \times n$ ”, to describe our desired output $m(n+1) = mn + m$. We thus define $(p_s(m)(n))(x) := add_{\mathbb{N}}(x, m)$, by which we really mean $m : \mathbb{N} \vdash mulsucc_{\mathbb{N}}(m) := \lambda n. \lambda x. add_{\mathbb{N}}(x, m) : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$.

b

One way we could define $exp_{\mathbb{N}} : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ is by

$$exp_{\mathbb{N}}(m)(0_{\mathbb{N}}) := 1 \equiv succ_{\mathbb{N}}(0_{\mathbb{N}})$$

$$exp_{\mathbb{N}}(m)(succ_{\mathbb{N}}(n)) := mul_{\mathbb{N}}(m)(exp_{\mathbb{N}}(m)(n))$$

Conveying the usual properties of exponentiation $m^0 = 1$ and $m^{n+1} = m^n \cdot m$. Proceeding in the same way as the previous part we define $m : \mathbb{N} \vdash expsucc_{\mathbb{N}}(m) := \lambda n. \lambda x. mul_{\mathbb{N}}(m)(x) : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$.¹

¹Note the order of arguments for the mul and add terms in the above matter in the sense that when things are unwrapped definitionally we get either 3+3+3+3 or 4+4+4 for 4*3 or 3*4, distinctly, and the same kind of thing for multiplication, based on what variable we inducted on, so to make our lives easier in checking properties and whatnot we care about the order.

2

We will define a max function $max_{\mathbb{N}} : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ while the min function definition should be clear as a parallel to this construction.

Intuitively we want to perform a double induction to encompass the following usual relations of a max:

$$max_{\mathbb{N}}(0_{\mathbb{N}}, 0_{\mathbb{N}}) := 0_{\mathbb{N}}$$

$$max_{\mathbb{N}}(0_{\mathbb{N}}, succ(j)) := succ(j)$$

$$max_{\mathbb{N}}(succ(k), 0_{\mathbb{N}}) := succ(k)$$

$$max_{\mathbb{N}}(succ(i), succ(l)) := succ(max_{\mathbb{N}}(i, l))$$

We do this by constructing an induction step as a term of type $\mathbb{N} \rightarrow ((\mathbb{N} \rightarrow \mathbb{N}) \rightarrow (\mathbb{N} \rightarrow \mathbb{N}))$, where we interpret our first input as $n : \mathbb{N}$ (our first term in our context), our second as $max_{\mathbb{N}}(n, -)$ (our second term in our context, call this f_n for convenience), and our output as $max_{\mathbb{N}}(succ(n), -)$.

We construct our desired output by using N-IND again, with our inductive contexts being $p'_0 := succ(n)$, $p'_s : \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ by $p'_s(j)(x) := succ(f_n(j))$. We then apply N-IND again to finish.

More explicitly, to construct $p_s(m)$ (our inductive step for the final construction) we construct

$$\lambda n. \lambda f_n. \text{N-IND}((\lambda r. \mathbb{N})(succ(n))(\lambda j. \lambda x. succ(f_n(j))))$$

Where we are saying in the context n, f_n , with our $P'(n)$ type just being constant $\mathbb{N}$, we have our base case term as $succ(n)$ (being $max_{\mathbb{N}}(succ(n), 0_{\mathbb{N}})$) and given $j : \mathbb{N}$ along with $x : \mathbb{N}$ (which we interpret as $max_{\mathbb{N}}(succ(n), j)$) we can write $max_{\mathbb{N}}(succ(n), succ(j))$ as just $succ(max_{\mathbb{N}}(n, j))$, which further is just $succ(f_n(j))$.

This construction gives us a way to describe our inductive step in the original setting, i.e. how to get $f_{succ(n)} : \mathbb{N} \rightarrow \mathbb{N}$ from $f_n : \mathbb{N} \rightarrow \mathbb{N}$, which we can then use, applying N-IND again, to get our full max function.

For $min_{\mathbb{N}}$ the key thing to notice is that $min_{\mathbb{N}}(succ(n), succ(j))$ should be the same as $succ(min_{\mathbb{N}}(n, j))$ and essentially the only thing that changes are what our base cases are.

3

a

We consider a constant type family $\mathbb{N}$ over $\mathbb{N}$. Our base case is defined as $0_{\mathbb{N}}$ and our inductive step, of type $\mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$, is defined by $\lambda n. \lambda x. add_{\mathbb{N}}(x, succ(n))$, just conveying that the $n + 1$ st triangular number is the previous one, x , summed with $n + 1$. We then apply N-IND.

b

We consider a constant type family $\mathbb{N}$ over $\mathbb{N}$. Our base case is defined by $\text{succ}(0_{\mathbb{N}})$ and our inductive step, of type $\mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$, is defined by $\lambda n.\lambda x.\text{mul}_{\mathbb{N}}(x, \text{succ}(n))$, just conveying that $(n+1)! = n!(n+1)$. We then apply N-IND.